Kernix 0.9.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

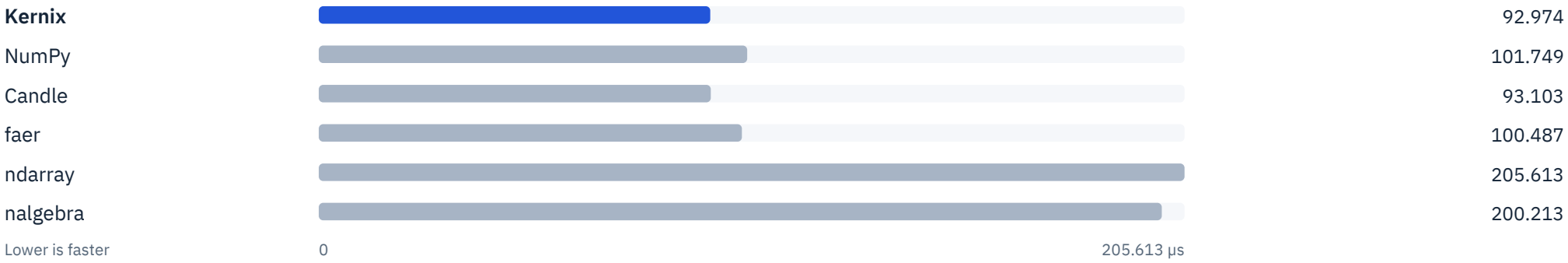
25 workloads

100 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

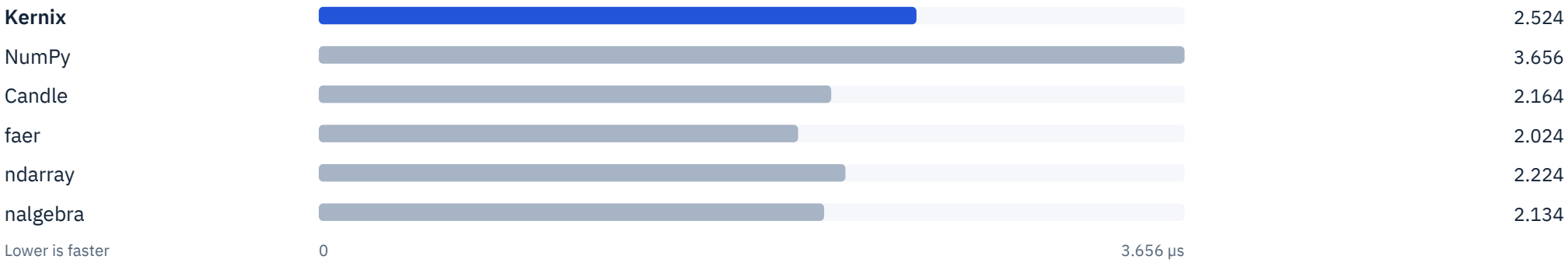
Kernix 0.9.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

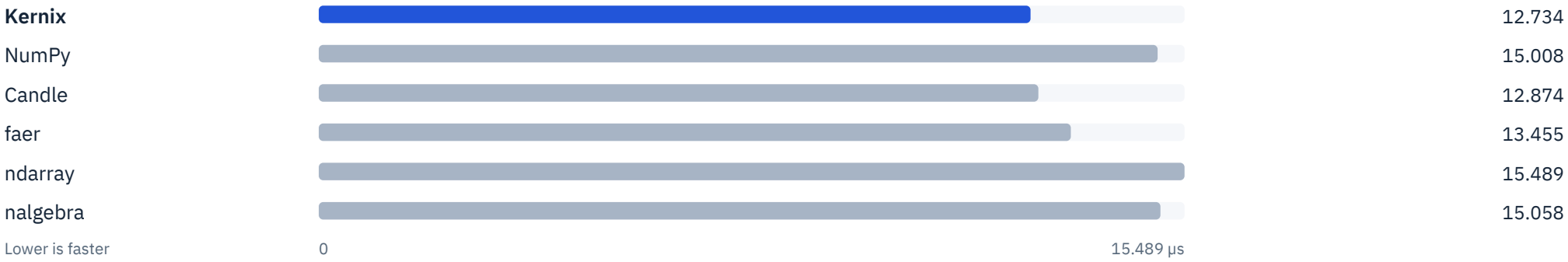
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



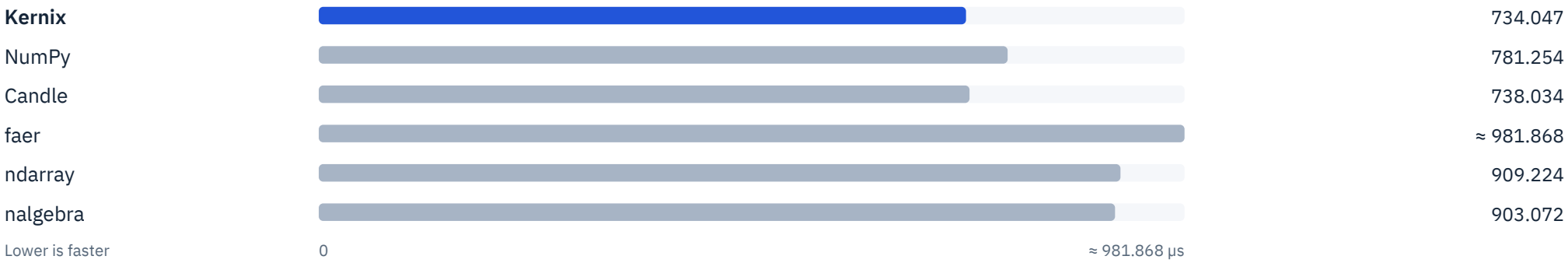
Kernix 0.9.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

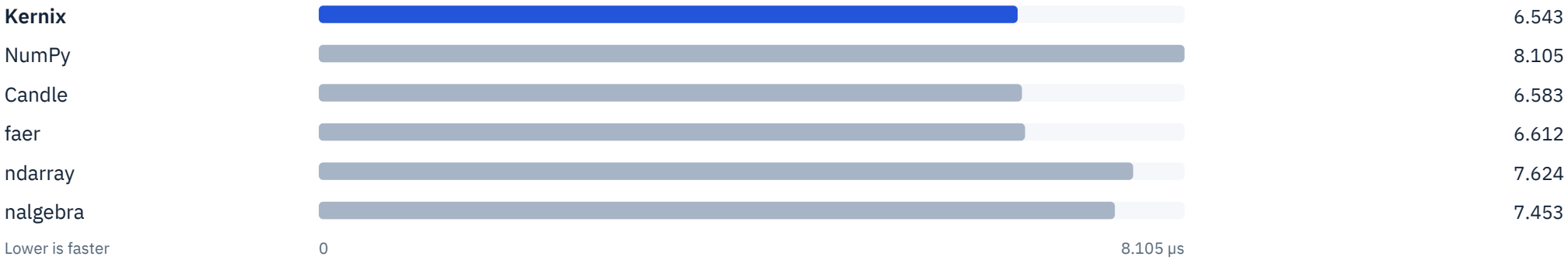
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



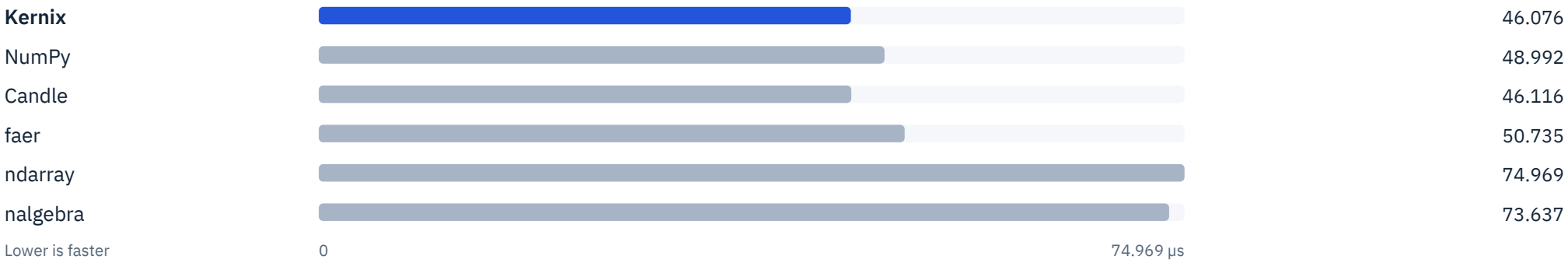
Kernix 0.9.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

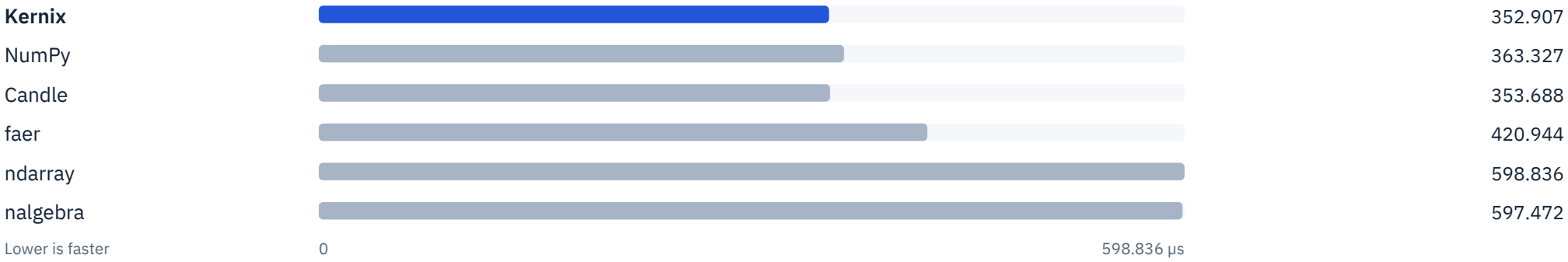
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.9.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

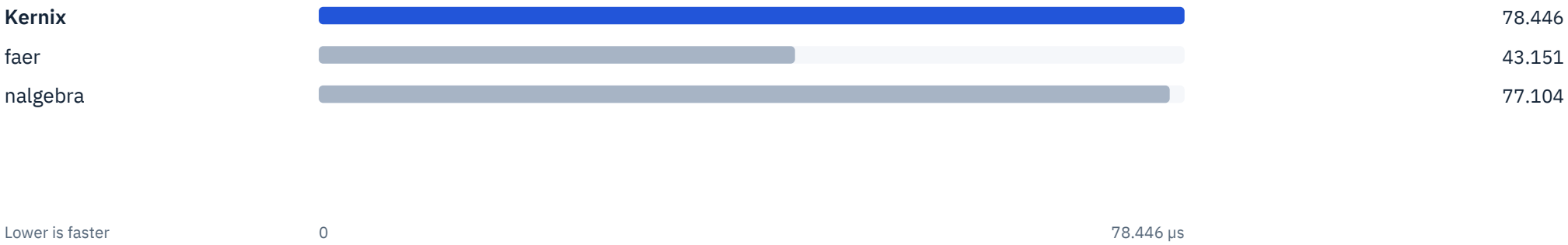


Kernix 0.9.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

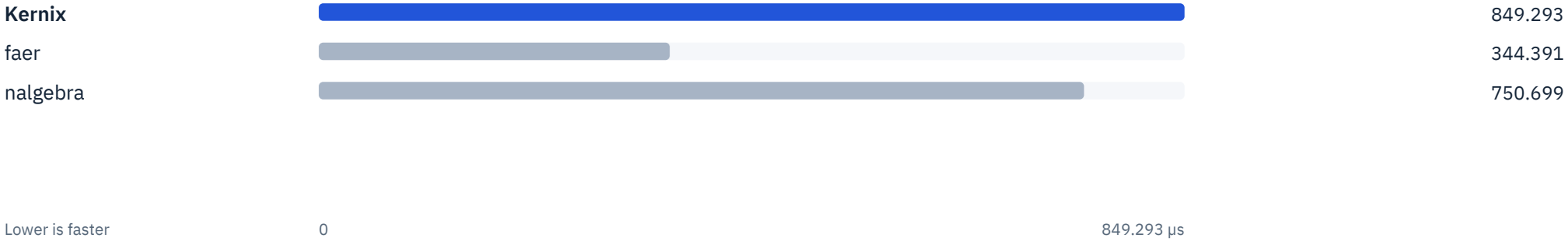
Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output



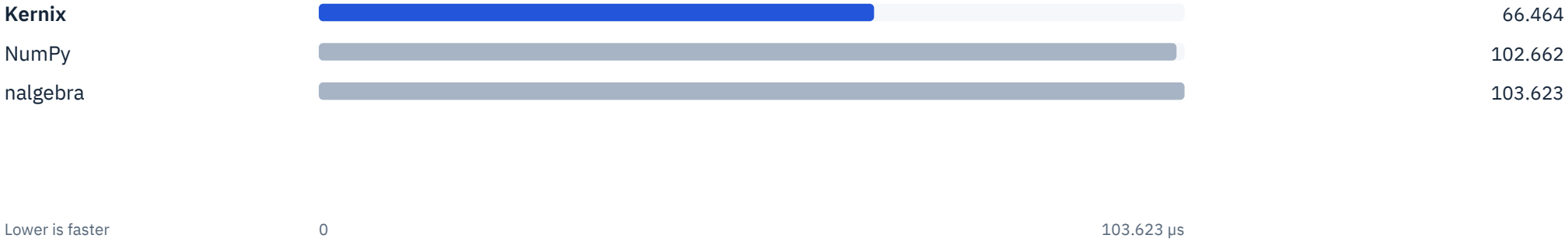
Kernix 0.9.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

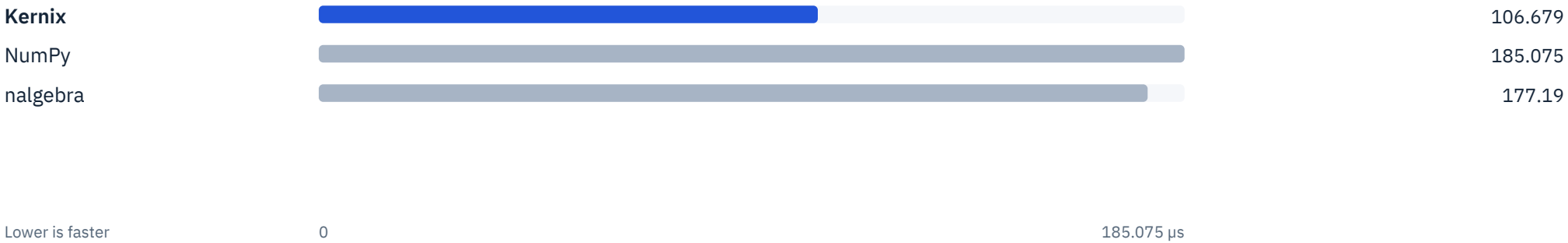
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

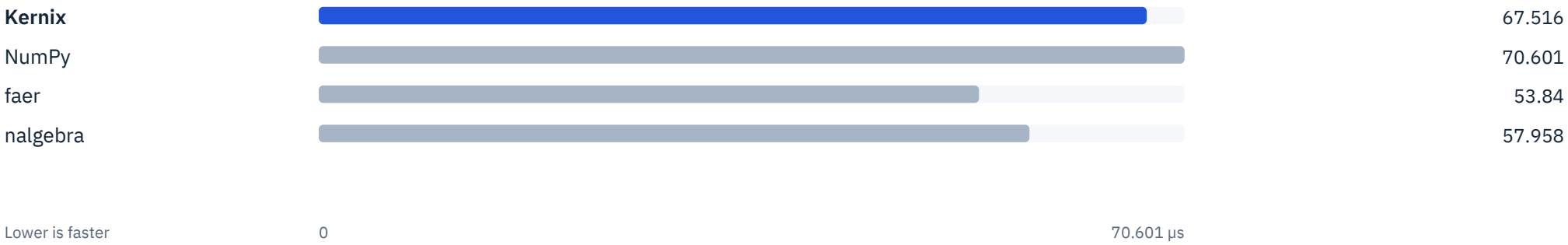


Kernix 0.9.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

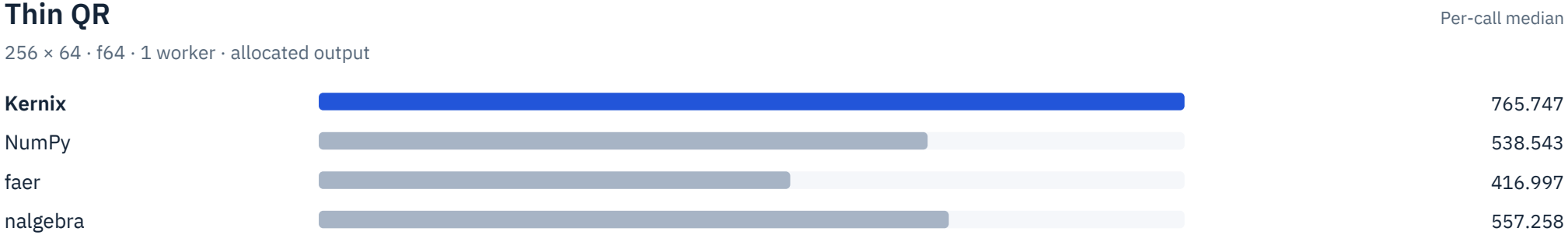
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



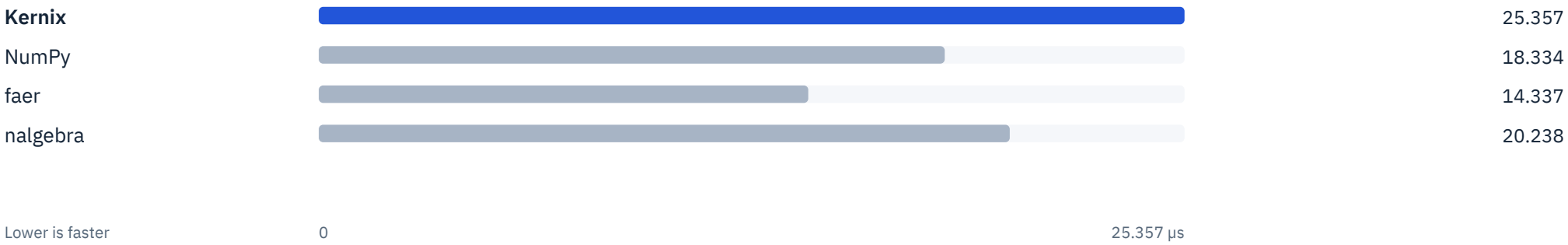
Kernix 0.9.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

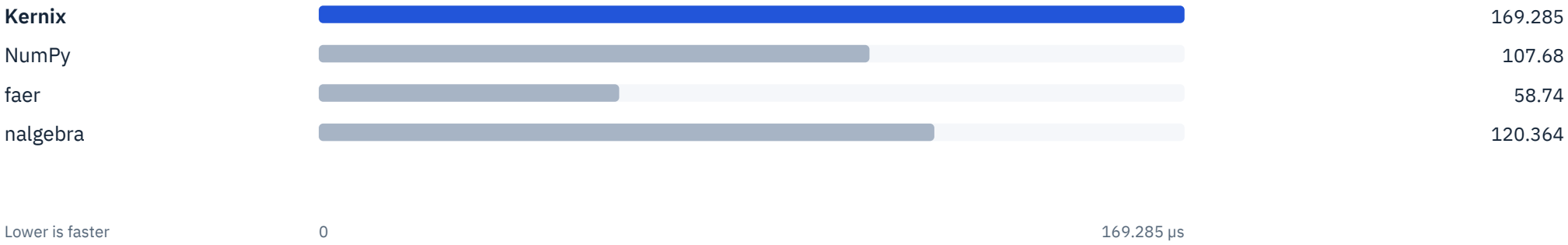
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



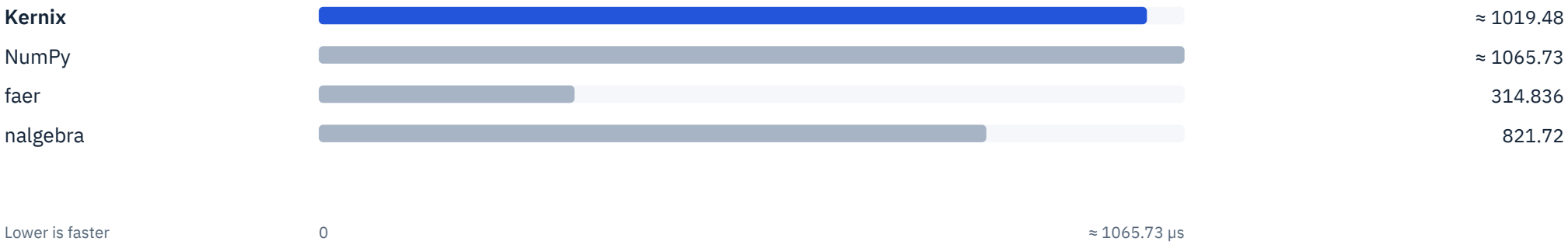
Kernix 0.9.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.9.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

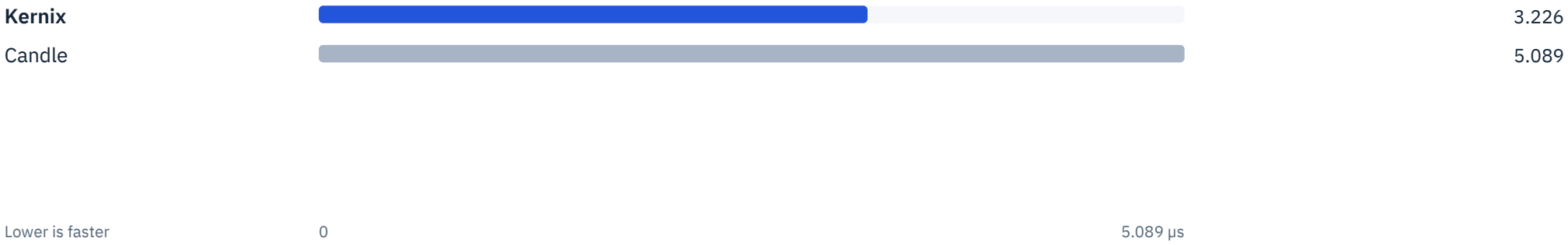
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.9.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

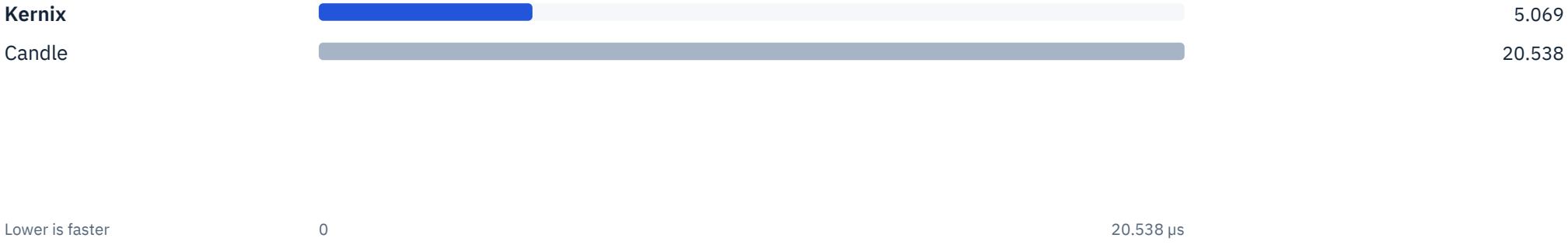
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.9.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

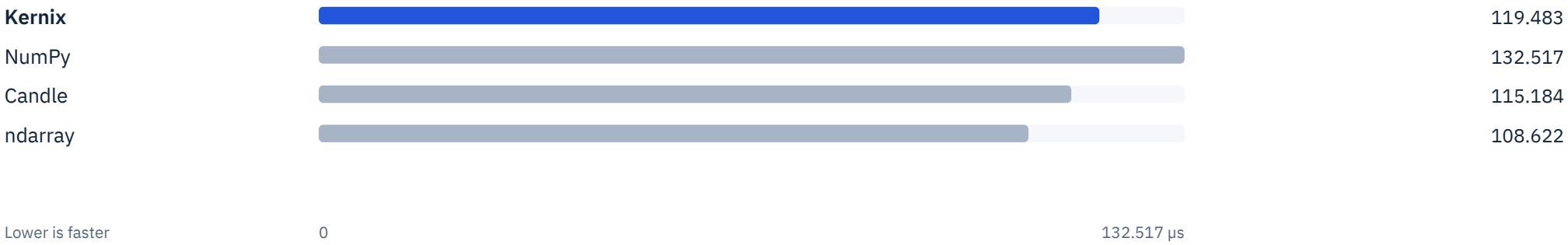


Kernix 0.9.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

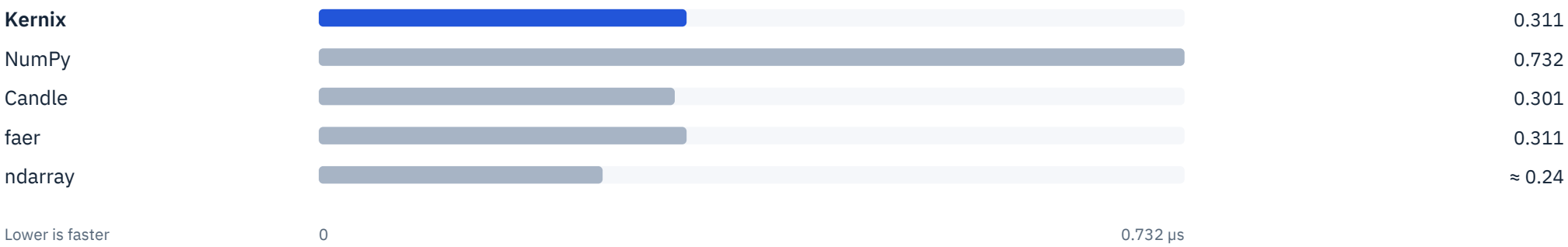


Kernix 0.9.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2524	2.524
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2024	2.024
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2224	2.224
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2164	2.164
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	3656	3.656
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2134	2.134
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	15008	15.008
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	15058	15.058
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12734	12.734
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	15489	15.489

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	13455	13.455
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	12874	12.874
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	100487	100.487
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	101749	101.749
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	200213	200.213
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	205613	205.613
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	92974	92.974
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	93103	93.103
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	903072	903.072
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	781254	781.254

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	734047	734.047
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	738034	738.034
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	981868.0000000001	981.8680000000001
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	909224	909.224
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7453	7.453
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	8105	8.105
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	6583	6.583
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6543	6.543
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	6612	6.612
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7624	7.624

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	46076	46.076
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	48992	48.992
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	73637	73.637
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	46116	46.116
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	74969	74.969
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	50735	50.735
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	363327	363.327
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	420944	420.944
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	352907	352.907
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	597472	597.472

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	353688	353.688
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	598836	598.836
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3246	3.246
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	4308	4.308
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3106	3.106
Column-pivoted QR / faer 96 × 32 · f64	43151	43.151
Column-pivoted QR / Kernix 96 × 32 · f64	78446	78.446
Column-pivoted QR / nalgebra 96 × 32 · f64	77104	77.104
Column-pivoted QR / nalgebra 256 × 64 · f64	750699	750.699
Column-pivoted QR / faer 256 × 64 · f64	344391	344.391

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / Kernix 256 × 64 · f64	849293	849.293
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	66464	66.464
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	102662	102.662
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	103623	103.623
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	106679	106.679
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	185075	185.075
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	177190	177.19
Thin QR / faer 96 × 32 · f64	53840	53.84
Thin QR / NumPy 96 × 32 · f64	70601	70.601
Thin QR / nalgebra 96 × 32 · f64	57958	57.958

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 96 × 32 · f64	67516	67.516
Thin QR / faer 256 × 64 · f64	416997	416.997
Thin QR / Kernix 256 × 64 · f64	765747	765.747
Thin QR / nalgebra 256 × 64 · f64	557258	557.258
Thin QR / NumPy 256 × 64 · f64	538543	538.543
Cholesky factorization / Kernix 64 × 64 · f64	25357	25.357
Cholesky factorization / nalgebra 64 × 64 · f64	20238	20.238
Cholesky factorization / faer 64 × 64 · f64	14337	14.337
Cholesky factorization / NumPy 64 × 64 · f64	18334	18.334
Cholesky factorization / faer 128 × 128 · f64	58740	58.74

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	169285	169.285
Cholesky factorization / NumPy 128 × 128 · f64	107680	107.68
Cholesky factorization / nalgebra 128 × 128 · f64	120364	120.364
Cholesky factorization / nalgebra 256 × 256 · f64	821720	821.72
Cholesky factorization / faer 256 × 256 · f64	314836	314.836
Cholesky factorization / Kernix 256 × 256 · f64	1019479	1019.479
Cholesky factorization / NumPy 256 × 256 · f64	1065726	1065.726
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3977	3.977
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7514	7.514
Model inference / Kernix batched normalized/sliced MLP · f64	3226	3.226

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	5089	5.089
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4148	4.148
Model recording + forward / Candle batched normalized/sliced MLP · f64	6272	6.272
Model backward / Candle prepared graph → fresh gradients · f64	20538	20.538
Model backward / Kernix prepared graph → fresh gradients · f64	5069	5.069
Parameter update / Kernix transactional SGD, fixed gradients · f64	831	0.831
Row softmax / ndarray 256 × 64 · f64	108622	108.622
Row softmax / NumPy 256 × 64 · f64	132517	132.517
Row softmax / Candle 256 × 64 · f64	115184	115.184
Row softmax / Kernix 256 × 64 · f64	119483	119.483

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	311	0.311
Create zeros / faer 64 × 64 · f32	311	0.311
Create zeros / ndarray 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create zeros / NumPy 64 × 64 · f32	732	0.732
Create zeros / Candle 64 × 64 · f32	301	0.301
Create ones / faer 64 × 64 · f32	391	0.391
Create ones / Kernix 64 × 64 · f32	431	0.431
Create ones / Candle 64 × 64 · f32	471	0.471
Create ones / NumPy 64 × 64 · f32	2053	2.053
Create ones / ndarray 64 × 64 · f32	351	0.351